

PROMPT FOR RESOLVING THE KELVIN PROBLEM AND FINDING THE THREE-DIMENSIONAL HONEYCOMB OPTIMIZER

PIATRA INSTITUTE

ABSTRACT. This document contains a full frontier-model research prompt for determining the least-area partition of three-dimensional Euclidean space into equal-volume cells. It treats the Weaire–Phelan foam as the leading candidate, not as an assumption, and requires a global proof, a rigorously certified better optimizer, or a proof that the infimum is not attained. The document also records the current mathematical status, the relation to higher-dimensional honeycomb problems and vector quantization, and the most promising proof-and-computation architecture. Status snapshot: 21 July 2026.

1. Prompt

Current task statement

A locally finite Caccioppoli partition of $\mathbb{R}^3$ is a countable family $\mathcal{E} = (E_i)_{i \in I}$ of measurable sets of locally finite perimeter such that the sets are pairwise disjoint up to null sets and cover $\mathbb{R}^3$ up to a null set. Every cell must have unit volume:

$$|E_i| = 1 \quad \text{for every } i \in I.$$

No periodicity, convexity, connectedness, congruence, bounded-diameter, fixed topology, or equal-pressure assumption is permitted unless it is proved to follow from the variational problem.

Let $J_{\mathcal{E}}$ be the union of the reduced interfaces, counted once:

$$J_{\mathcal{E}} = \bigcup_{i < j} (\partial^* E_i \cap \partial^* E_j).$$

For $Q_R = [-R/2, R/2]^3$, define the normalized asymptotic mean cell area

$$\mathcal{A}(\mathcal{E}) = \limsup_{R \rightarrow \infty} \frac{2 \mathcal{H}^2(J_{\mathcal{E}} \cap Q_R)}{|Q_R|}.$$

The factor 2 reports the full boundary area of a typical unit-volume cell rather than counting each physical film only once. For a periodic partition of the flat torus $\mathbb{T}_{\Lambda}^3 = \mathbb{R}^3/\Lambda$ into N unit-volume cells, this agrees with

$$\mathcal{A}(\mathcal{E}) = \frac{1}{N} \sum_{i=1}^N \text{Per}(E_i; \mathbb{T}_{\Lambda}^3).$$

Define the three-dimensional honeycomb constant

$$\kappa_3 = \inf_{\mathcal{E}} \mathcal{A}(\mathcal{E}),$$

where the infimum ranges over every admissible unit-volume partition above.

Resolve the Kelvin problem completely:

Determine κ_3 , prove whether the infimum is attained, identify at least one global optimizer when it is attained, and decide whether the relaxed Weaire-Phelan A15 foam is a global optimizer.

The Weaire-Phelan candidate is the periodic A15 foam with a cubic fundamental cell containing eight equal-volume regions: two cells with twelve pentagonal faces and six cells with twelve pentagonal and two hexagonal faces, after the interfaces have been relaxed under the individual volume constraints. The actual candidate has curved constant-mean-curvature films and Plateau singularities; the flat-faced polyhedral honeycomb and a finite Surface Evolver mesh are only approximations. Commonly reported normalized areas are approximately 5.288 for Weaire-Phelan and 5.306 for Kelvin. Recompute and rigorously enclose all benchmark values rather than treating these decimals as exact data.

Complete-resolution standard

A complete resolution must establish one of the following alternatives.

- **Weaire-Phelan optimality.** Give a mathematically precise construction or characterization of the relaxed Weaire-Phelan partition, prove exact unit volumes and admissibility, prove a rigorous upper bound for its area, and prove a universal lower bound for every admissible partition that matches it. The proof may be computer-assisted, but every numerical step must have a checkable error certificate. At least one equality case must be identified rigorously.
- **A different optimizer.** Construct an admissible partition with rigorously certified area, prove that it attains the universal infimum, and show whether it improves on Weaire-Phelan. A better numerical candidate without a matching global lower bound disproves Weaire-Phelan optimality but does not by itself resolve the Kelvin problem.
- **Nonattainment.** Prove that no admissible partition attains κ_3 , determine the infimum by matching upper and lower bounds, and characterize a minimizing sequence sufficiently to exclude hidden attainment. Do not assume a minimizer exists merely because finite-torus subproblems have minimizers.

A closed elementary expression for κ_3 is not required if the optimizer is defined exactly as the unique solution of a rigorously posed analytic or variational system and its value is enclosed by certified intervals that can be narrowed arbitrarily. A decimal obtained from floating-point descent is not an exact characterization.

What does not count

Partial progress is insufficient unless it logically implies the exact resolution above. In particular, the following do not count as a complete solution:

- showing only that Weaire-Phelan beats Kelvin, or finding another structure that merely beats Kelvin;
- showing that a candidate is stationary, satisfies Plateau's laws, is linearly stable, or is a local minimum;
- optimizing only the polyhedral A15 honeycomb, a fixed triangulated mesh, or a fixed Surface Evolver discretization;

- proving optimality only among Voronoi cells, power diagrams, lattice fundamental domains, congruent cells, convex cells, Frank-Kasper phases, a bounded list of topologies, or periodic cells with at most a fixed number of regions, without a theorem reducing the unrestricted problem to that class;
- imposing equal pressure instead of individual equal-volume constraints, or silently assuming that equal volumes force equal Lagrange multipliers;
- comparing finite foam clusters in a container without a boundary-error theorem for the infinite-volume limit;
- using a phase-field, threshold-dynamics, finite-element, or neural representation without a convergence theorem and a posteriori error bounds;
- an empirical search that fails to find a better structure, even after an enormous computation;
- a universal lower bound that remains strictly below the certified candidate area;
- transferring a conclusion from vector quantization, centroidal Voronoi tessellations, sphere packing, or a different cell energy without an exact implication for perimeter;
- a proof for the periodic problem alone unless periodicity of a global optimizer is proved or a periodic-approximation theorem closes the gap to the unrestricted problem.

Multiagent search protocol

Use multiagent research aggressively and dynamically. If up to 64 concurrent agents are available, do not use a fixed assignment such as “N agents for strategy X.” Manage the search using the following rules:

- Begin with a genuinely diverse portfolio of approaches. Preserve independence in early rounds so that agents do not all converge to the same attractive numerical or geometric picture.
- Maintain an explicit registry of approach families, key lemmas, computed artifacts, assumptions, failure modes, and exact remaining gaps. Group routes by mathematical mechanism, not by wording.
- Keep several incompatible proof routes alive through multiple rounds. Cross-pollinate only after independent agents have developed them far enough to expose their actual strengths and defects.
- Mark a route as blocked when it reaches a missing lemma equivalent in strength to the original problem. Reopen it only when a new invariant, dual certificate, compactness mechanism, topological reduction, or exact computation is proposed.
- Use adversarial agents throughout. Every proposed theorem, calibration, numerical certificate, topology reduction, and periodic-limit argument must be attacked by agents that did not originate it.
- Require concrete lemmas, equations, meshes, code, interval certificates, counterexamples to sublemmas, or formal statements. Reject status reports, vague optimism, visual evidence, and claims that a global compatibility step is “routine.”
- The root agent must repeatedly synthesize, challenge, redirect, and launch new rounds. Do not stop after the first portfolio fails.
- Use public literature search actively for background, reductions, prior candidates, and novelty auditing. Never treat a literature statement, simulation figure, or community consensus as a proof.

Problem-specific search portfolio

The initial independent rounds must include, but are not limited to, the following families. Regardless of whether the unrestricted problem is resolved, treat determination of the

continuum minimizer within the full A15 topological class as a mandatory intermediate benchmark: optimize the lattice, interfaces, pressures, and all topology-preserving degrees of freedom, then certify the resulting area. This benchmark is not by itself a solution of the Kelvin problem.

1. **Formalization and normalization.** Prove that the chosen asymptotic functional is well posed; compare limsup, Følner-sequence, stationary-random, and periodic formulations; track whether interfaces are counted once or twice; and isolate the exact hypotheses needed for compactness and existence.
2. **Certified Weaire-Phelan reconstruction.** Start from the public A15/Surface Evolver combinatorics, independently rebuild the eight-cell torus, optimize the lattice as well as the films, enforce all eight volumes with one redundant constraint removed, and compute validated intervals for volume, area, mean curvature, Plateau angles, and residuals. Produce a mesh-refinement convergence study and a representation suitable for rigorous numerics.
3. **Optimizer-first A15 track.** Determine the true area-minimizing partition within the complete A15 topological class, not merely within one mesh. Test cubic and non-cubic lattice deformations, possible pressure splitting between the two cell types, symmetry breaking, and all admissible topology-preserving perturbations. Prove local minimality or find a descent direction using the constrained second variation and the Jacobi operator.
4. **Paired calibrations and convex duality.** Search for periodic vector fields or differential forms ξ_i satisfying a global inequality such as

$$|\xi_i - \xi_j| \leq 1$$

throughout the torus, constant or otherwise controlled divergences compatible with the volume constraints, and equality of the normal trace on every candidate interface. A valid paired calibration would convert the area comparison against all competitors into a divergence-theorem identity and is a priority route because it can provide a global lower bound and certify equality simultaneously. Explore generalized calibrations, stress tensors, convex duals of total variation, and interval-certified PDE constructions if classical smooth fields are too rigid.

5. **Hales-style local inequalities.** Seek a decomposition of global area into local contributions from cells, faces, Plateau borders, and vertices. Combine Euler relations, Gauss-Bonnet terms, spherical links, mean-curvature identities, face-degree distributions, isoperimetric deficits, and topological charge. The target is a sharp inequality whose equality conditions reproduce A15 or reveal a better topology, not merely an averaged face-count heuristic.
6. **Topological and combinatorial enumeration.** Represent periodic Plateau foams by cell complexes on a three-torus and by dual triangulations. Use exact graph generation, SAT/SMT, integer programming, and symmetry reduction to enumerate or constrain low-complexity competitors. Include A15, C15, Z, sigma, other tetrahedrally close-packed structures, non-Frank-Kasper complexes, and topological mutations. Any finite enumeration must be paired with a theorem proving why excluded complexities cannot win.
7. **Global computational discovery.** Run independent Surface Evolver, finite-element, level-set, multiphase-field, threshold-dynamics, and differentiable-cell-complex searches. Vary the number of cells per fundamental domain, the lattice, initial point sets, topology, and symmetry. Include large periodic approximants to quasiperiodic and disordered foams. Use computation to discover candidates and lemmas, not as a substitute for proof.
8. **Aperiodic and thermodynamic-limit analysis.** Determine whether periodic partitions are dense in energy among all admissible partitions. Attempt a cut-and-paste periodic approximation theorem with explicit boundary cost, or construct a mechanism by which quasiperiodicity or disorder can lower the area. Prove diameter, inradius, neighbor-count, or local-complexity bounds if they are needed to prevent loss of compactness.

9. **Regularity and singular structure.** Use geometric measure theory for minimizing clusters to justify constant mean curvature on interfaces, 120° triple lines, tetrahedral four-line vertices, and the absence of additional singularities in dimension three. Do not assume that necessary Plateau conditions imply global minimality.
10. **Rigorous numerics and formal verification.** Use rational arithmetic, interval arithmetic, validated finite elements, exact topology files, and independently written checkers. Formalize finite combinatorics and delicate inequality chains in Lean, Isabelle, or another proof assistant when practical. A computer-assisted proof must be reproducible on commodity hardware or supply a verifiable certificate whose checking cost is substantially lower than its discovery cost.
11. **Higher-dimensional transfer.** Define

$$\kappa_d = \inf_{\mathcal{E}} \limsup_{R \rightarrow \infty} \frac{\sum_i \text{Per}(E_i; Q_R)}{|Q_R|}$$
 for unit-volume partitions of $\mathbb{R}^d$. Search for dimension-independent inequalities, product constructions, induction obstructions, and candidate families, but do not allow the broader program to dilute the primary requirement of resolving $d = 3$.
12. **Quantization and crystallization as controlled analogies.** Compare the perimeter objective with normalized second moment, centroidal Voronoi energy, block-copolymer free energies, and Frank-Kasper crystallization. Extract transferable methods and candidate complexes while proving every bridge. In particular, do not infer perimeter optimality from quantizer optimality or vice versa.

Priority mechanism: a periodic paired calibration

Treat the following as the highest-value proof experiment, not as an assumption that it must succeed.

For a rigorously reconstructed candidate partition on an eight-cell torus, search for fields $\xi_1, \dots, \xi_8$ such that, with the correct orientation conventions,

$$|\xi_i - \xi_j| \leq 1 \quad \text{everywhere,} \quad (\xi_i - \xi_j) \cdot \nu_{ij} = 1 \quad \text{on each interface } \Sigma_{ij},$$

and the divergence terms reduce to constants fixed by the prescribed volumes. Exploit A15 symmetry to reduce the PDE or optimization variables. Parameterize the fields by periodic Fourier bases, finite elements, harmonic extensions, or solutions of elliptic equations driven by the candidate normals. Optimize the global slack in the norm inequalities, then convert any positive numerical slack into interval-certified inequalities. If no classical paired calibration can exist, prove an obstruction and use it to design a generalized calibration rather than abandoning the dual route.

Mandatory adversarial audit

Every surviving candidate resolution must be checked against all items below by independent agents:

- exact unit volumes, torus periods, periodic wraps, and interface orientations;
- the factor-of-two convention between physical film area and mean cell boundary area;
- curved films rather than their flat polyhedral proxies;
- all Plateau borders and vertices, including possible hidden singularities created by topology changes;
- individual volume constraints and the resulting pressure multipliers;
- boundary terms in finite clusters, large boxes, and periodic approximation arguments;
- mesh dependence, discretization error, convergence rate, and conditioning of near-zero Hessian modes;

- whether a claimed universal inequality secretly assumes convexity, congruence, periodicity, bounded face number, or a fixed topology;
- whether equality conditions actually characterize the proposed optimizer;
- independent reproduction of every numerical upper bound, lower bound, and certificate;
- novelty against the complete Kelvin, foam, minimal-cluster, calibration, Frank-Kasper, phase-field, and quantization literature.

Required deliverables for a claimed solution

Return all of the following, not merely a prose proof:

1. a precise theorem statement in the admissible class above;
2. a complete proof with every external theorem stated in the exact form used;
3. a machine-readable description of the optimizer or minimizing sequence;
4. source code, build instructions, fixed random seeds, input data, and exact versions of all solvers;
5. independently checkable certificates for topology, volume, area, inequalities, and numerical error;
6. an adversarial audit report listing every attempted falsification and its result;
7. a novelty report distinguishing new mathematics from known lemmas and standard computations.

If a complete proof is found, first have multiple independent agents reconstruct it from the theorem statement and artifacts alone. Then have a separate team attempt to produce an explicit lower-area competitor or a counterexample to each key lemma.

Termination rule

Do not return merely because the problem is known to be open, because initial numerical searches reproduce Weaire-Phelan, or because agents encounter theorem-strength gaps. Continue launching new rounds and reopening blocked routes only when a materially new mechanism appears. Carry out at least twelve full synthesis-and-redirection rounds and spend at least eight hours of sustained research before considering a final response.

Return a claimed resolution only when the global theorem and all certificates survive adversarial audit. If the execution environment forcibly ends before that point, return a clearly labeled non-solution dossier containing only rigorously established new lemmas, certified bounds, reproducible candidates, failed routes, and the exact remaining gap. Never present numerical plausibility as a proof.

2. Current Status and the Point of Attack

The two-dimensional equal-area perimeter problem is solved: Hales proved that the regular hexagonal honeycomb minimizes perimeter among general planar partitions. The corresponding unrestricted problem in three dimensions remains open. Weaire and Phelan’s 1994 A15 construction numerically improved Kelvin’s foam by roughly 0.3% and remains the standard best-known candidate, but no universal lower bound is close enough to certify it. Numerical studies of periodic, Frank–Kasper, and quasiperiodic competitors have found structures near the Weaire–Phelan value rather than a proof of global optimality.[1, 2, 8]

Recent rigorous work is active mainly on existence, compactness, regularity, constrained periodic variants, and numerical approximation. Cesaroni and Novaga proved existence for periodic equal-cell anisotropic problems and, in a 2025 variant, existence and regularity when every cell contains a ball of fixed radius. Their discussion emphasizes that identification of the optimizer remains open even in important lattice-restricted formulations.[9, 10]

The gap between the sphere lower bound and the candidate is still substantial. A unit-volume sphere has area

$$(36\pi)^{1/3} \approx 4.835976,$$

whereas the reported Weaire–Phelan value is around 5.288. Thus a proof cannot rely on the single-cell isoperimetric inequality alone; it must quantify the unavoidable global frustration of fitting nearly spherical cells together.

Layer	What is currently available	Missing step
Candidate construction	Public A15 combinatorics, Surface Evolver files, repeated numerical relaxation	Rigorous continuum definition, certified area, and global comparison
Local geometry	Plateau laws and constant-mean-curvature conditions for regular minimizers	A theorem turning local stationarity into global optimality
Computation	Strong tools for surface descent, phase fields, finite elements, and topology search	Proof-producing error control and exhaustive reduction
Existence/compactness	Results for fixed or constrained periodic problems and generalized clusters	An unrestricted optimizer theorem or a precise nonattainment mechanism
Global lower bounds	Cellwise isoperimetry, topology identities, and general cluster theory	A sharp inequality or calibration reaching the A15 value

The most promising proof-shaped target is therefore not “search a larger catalogue of foams” but “construct a sharp dual certificate for one candidate.” A successful periodic paired calibration would prove a lower bound against every competitor in the same torus problem while attaining equality on the candidate. The remaining major bridge would then be a theorem connecting the torus problem to the unrestricted thermodynamic limit. This route is difficult, but it has the correct verification asymmetry: a proposed calibration can be checked independently and globally.

3. Higher Dimensions and Vector Quantization

A higher-dimensional honeycomb problem replaces surface area by $(d - 1)$ -dimensional perimeter and asks for κ_d for unit-volume partitions of $\mathbb{R}^d$. Beyond $d = 2$, general optimizers are unknown. Existing rigorous literature gives existence and qualitative results for selected periodic, anisotropic, or constrained formulations rather than a classification of optimal cells. In dimension four, the regular 24-cell honeycomb is a natural candidate for a fixed-inradius lattice formulation; recent work verifies a local perimeter-minimality

property and formulates a 24-cell conjecture, but this is not a solution of the unrestricted equal-volume problem.[9, 10]

The connection to vector quantization is real but easy to misstate. High-rate quantization minimizes a second-moment distortion of cells around codepoints, not their boundary area. Both problems favor uniform, nearly isotropic space partitions and therefore share Voronoi methods, crystallization phenomena, and candidate lattices. They need not have the same optimizer: Kashyap and Neuhoff showed that the Weaire–Phelan partition has a larger effective normalized moment of inertia than the truncated-octahedral BCC partition, even though it has lower surface area. The three-dimensional Gersho problem for optimal centroidal Voronoi tessellations also remains open.[11, 12]

This distinction suggests a productive multiobjective program rather than a claimed equivalence. One can study the Pareto frontier between normalized perimeter, normalized second moment, spectral cell energies, manufacturability, and mechanical or transport performance. A structure globally optimal for one functional may be a useful adversarial example for another.

4. Potential Applications

Domain	Mathematical connection	Realistic application
Dry foams and emulsions	Surface tension minimizes interfacial area subject to bubble volumes	Equilibrium microstructure, coarsening benchmarks, and constitutive models
Block copolymers and soft matter	Competition between interfacial area and chain-stretching frustration produces Frank–Kasper phases	Predicting and designing self-assembled A15, sigma, and related phases
Photonic materials	Foam networks provide isotropic three-dimensional dielectric architectures	Weaire–Phelan-derived designs have produced sizeable complete photonic band gaps in simulation
Heat and mass transfer	Cellular geometry controls area, pressure drop, tortuosity, and mixing	Additively manufactured heat exchangers, catalysts, and porous media, usually under multiobjective rather than pure-area optimization
Structural and architectural design	Periodic cellular complexes distribute material through space	Lightweight shells and space frames; the Kelvin optimizer is a geometric starting point, not automatically a structural optimum
Vector quantization and meshing	Both seek uniform, isotropic partitions, but with different energies	Codebook design, clustering, numerical integration, and high-quality volumetric meshes

The strongest scientific payoff would be broader than naming one foam. A proof would likely introduce a reusable method for sharp lower bounds in infinite multiphase geometric optimization. A counterexample would reveal a new principle of three-dimensional space partitioning and immediately redirect work in foams, crystallization, and cellular design.

5. References

References

- [1] W. Thomson (Lord Kelvin), “On the division of space with minimum partitional area,” *Philosophical Magazine* 24 (1887), 503–514. DOI: [10.1080/14786448708628135](https://doi.org/10.1080/14786448708628135).
- [2] D. Weaire and R. Phelan, “A counter-example to Kelvin’s conjecture on minimal surfaces,” *Philosophical Magazine Letters* 69 (1994), 107–110. DOI: [10.1080/09500839408241577](https://doi.org/10.1080/09500839408241577).

- [3] K. A. Brakke, “The Surface Evolver,” *Experimental Mathematics* 1 (1992), 141–165. DOI: [10.1080/10586458.1992.10504253](https://doi.org/10.1080/10586458.1992.10504253).
- [4] R. Phelan, D. Weaire, and K. Brakke, “Computation of equilibrium foam structures using the Surface Evolver,” *Experimental Mathematics* 4 (1995), 181–192. DOI: [10.1080/10586458.1995.10504320](https://doi.org/10.1080/10586458.1995.10504320).
- [5] T. C. Hales, “The honeycomb conjecture,” *Discrete & Computational Geometry* 25 (2001), 1–22. DOI: [10.1007/s004540010071](https://doi.org/10.1007/s004540010071).
- [6] E. Oudet, “Approximation of partitions of least perimeter by Γ -convergence: around Kelvin’s conjecture,” *Experimental Mathematics* 20 (2011), 260–270. DOI: [10.1080/10586458.2011.565233](https://doi.org/10.1080/10586458.2011.565233).
- [7] R. Gabbriellini, “A new counter-example to Kelvin’s conjecture on minimal surfaces,” *Philosophical Magazine Letters* 89 (2009), 483–491. DOI: [10.1080/09500830903022651](https://doi.org/10.1080/09500830903022651).
- [8] S. J. Cox, F. Graner, R. Mosseri, and J.-F. Sadoc, “Quasicrystalline three-dimensional foams,” *Journal of Physics: Condensed Matter* 29 (2017), 114001. DOI: [10.1088/1361-648X/aa5712](https://doi.org/10.1088/1361-648X/aa5712).
- [9] A. Cesaroni and M. Novaga, “Minimal periodic foams with equal cells,” in *Anisotropic Isoperimetric Problems and Related Topics*, Springer INdAM Series 62 (2024), 15–24. DOI: [10.1007/978-981-97-6984-1_2](https://doi.org/10.1007/978-981-97-6984-1_2).
- [10] A. Cesaroni and M. Novaga, “Minimal periodic foams with fixed inradius,” *Mathematika* 71 (2025), e70020. DOI: [10.1112/mtk.70020](https://doi.org/10.1112/mtk.70020).
- [11] N. Kashyap and D. L. Neuhoff, “On quantization with the Weaire–Phelan partition,” *IEEE Transactions on Information Theory* 47 (2001), 2538–2543. DOI: [10.1109/18.945264](https://doi.org/10.1109/18.945264).
- [12] R. Choksi and X. Y. Lu, “Bounds on the geometric complexity of optimal centroidal Voronoi tessellations in 3D,” *Communications in Mathematical Physics* 377 (2020), 2429–2450. DOI: [10.1007/s00220-020-03789-y](https://doi.org/10.1007/s00220-020-03789-y).
- [13] M. A. Klatt, P. J. Steinhardt, and S. Torquato, “Phoamtonic designs yield sizeable 3D photonic band gaps,” *Proceedings of the National Academy of Sciences* 116 (2019), 23480–23486. DOI: [10.1073/pnas.1912730116](https://doi.org/10.1073/pnas.1912730116).
- [14] J. Fischer, S. Hensel, T. Laux, and T. M. Simon, “Local minimizers of the interface length functional based on a concept of local paired calibrations,” arXiv:2212.11840 (2022).